

Index	Data Source	Changes
1	Scopus	First version based on PICOC criteria
2	Scopus	Removal of AND ("safety-critical" OR "real-time" OR "mixed-criticality" OR determinism OR "core affinity" OR "performance optimization")
3	Scopus	Removal of 'vehicle' OR ADAS. Addition of AND ("safety-critical" OR "real-time" OR "mixed-criticality" OR determinism OR "core affinity" OR "performance optimization")
4	Scopus	mixed keywords removed
5	Scopus	Search string split into two research questions: Core types and properties and allocation methods
6	Scopus	Addition of "OR propert*"
7	Scopus	removal of OR "runtime management" OR "load balancing". "OR software-defined vehicle"

8	Scopus	Add " OR UAV " exclusion
9	Scopus	Final search string. Removed vehicle, software-defined vehicle and UAV condition to keep search string lean

Search String Evolution Rapid Review RunSoC2.0

Status as of 14.05.2026

Search String
TITLE-ABS-KEY((automotive OR vehicle OR "software-defined vehicle" OR ADAS) AND ("System-on-Chip" OR SoC OR "heterogeneous multicore" OR "heterogeneous processor" OR "heterogeneous platform" OR "multi-core architecture" OR MPSoC) AND (allocation OR mapping OR scheduling OR "task allocation" OR "software allocation" OR "runtime management" OR "load balancing") AND ("safety-critical" OR "real-time" OR "mixed-criticality" OR determinism OR "core affinity" OR "performance optimization") AND NOT (battery OR charging OR powertrain-only))
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Effect of Update	Number of Results
Original	145
Large increase from broadened search	315
Large reduction. Increase in relevance	62
Large increase	111
Slight reduction	93
Large increase. No effect on relevance	184
None	107

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